

ADRAR, Algeria

WMO: 606200

Lat: 27.838N

Lon: 0.186W

Elev: 280

StdP: 98.01

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.1	4.8	-14.9	1.1	13.4	-13.0	1.3	14.8	15.3	15.4	13.9	15.6	4.6	30	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	47.1	22.3	46.2	21.9	45.3	21.6	23.6	40.5	22.8	40.2	21.9	40.2	5.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.0	31.2	16.9	12.5	31.4	15.1	11.1	32.5	71.8	40.6	68.4	40.2	65.1	40.1	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.8	11.4	DB	0.4	48.4	1.6	0.6	-0.8	48.9	-1.7	49.3	-2.6	49.6	-3.7	50.1	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	13.4	16.2	20.6	25.2	30.1	34.6	37.8	36.8	33.2	26.9	19.1	14.3
	DBStd	9.03	2.79	3.42	3.68	3.39	3.42	2.37	1.86	1.87	2.91	3.30	3.44	2.77
	HDD10.0	7	4	1	0	0	0	0	0	0	0	0	0	2
	HDD18.3	407	156	76	18	1	0	0	0	0	0	0	29	128
	CDD10.0	5754	108	174	329	457	624	739	861	832	697	522	274	136
	CDD18.3	3112	2	15	89	207	366	489	603	573	447	264	53	3
	CDH23.3	52329	51	272	1299	2929	5819	8684	11288	10464	7361	3424	666	71
CDH26.7	36706	3	63	573	1654	3878	6388	8816	7996	5143	1952	231	8	
Wind	WSAvg	6.0	5.6	6.1	6.3	6.5	6.4	6.0	6.3	6.1	5.6	5.6	5.4	5.6
Precipitation	PrecAvg	20	2	2	1	5	1	1	0	1	2	2	1	0
	PrecMax	84	20	19	8	83	7	6	2	15	18	24	19	11
	PrecMin	2	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	20	5	4	2	17	2	1	0	3	4	6	4	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.0	31.1	37.0	41.0	44.1	47.0	48.2	47.2	45.1	41.2	34.8	27.8
		MCWB	12.5	14.1	17.0	17.1	20.3	21.6	22.3	21.4	21.5	20.3	16.4	14.2
	2%	DB	24.8	28.2	34.1	38.2	42.8	45.8	47.7	46.2	44.1	38.8	31.8	25.1
		MCWB	11.8	13.0	15.6	18.0	19.9	21.4	22.3	21.7	21.4	19.3	16.1	13.2
	5%	DB	23.0	26.4	32.0	36.2	41.2	44.8	47.0	45.8	43.1	37.0	29.2	23.5
		MCWB	10.9	12.4	14.9	17.3	19.3	21.2	22.3	21.7	21.1	18.9	15.2	12.5
	10%	DB	21.2	24.8	30.0	34.5	39.8	43.3	46.0	44.9	41.9	35.1	27.1	21.9
		MCWB	10.2	11.7	14.1	16.6	19.2	20.6	21.9	21.5	20.9	18.2	14.6	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	15.8	18.1	21.1	22.2	22.9	24.1	25.7	25.0	22.8	18.9	16.4
		MCDB	23.6	25.3	26.2	31.2	37.9	41.3	44.4	41.7	38.4	35.6	28.6	24.4
	2%	WB	12.7	14.0	16.1	19.0	20.4	21.9	23.2	23.6	23.4	21.1	17.3	14.3
		MCDB	22.3	25.0	28.3	31.4	36.8	41.5	43.9	39.6	38.1	34.1	28.8	22.9
	5%	WB	11.6	12.8	14.9	17.3	19.3	20.9	22.2	22.6	22.4	19.9	16.0	13.1
		MCDB	21.6	24.1	27.6	31.3	35.9	40.3	43.7	39.9	37.5	33.3	27.1	22.1
	10%	WB	10.5	11.9	13.8	16.0	18.3	20.0	21.2	21.6	21.4	19.0	14.8	12.0
		MCDB	20.5	23.1	26.9	30.1	35.4	39.9	43.2	40.8	36.8	32.7	25.7	20.9
Mean Daily Temperature Range	5% DB	MDBR	15.4	15.6	16.0	16.0	16.1	16.4	16.4	15.5	15.0	14.8	14.9	14.9
		MCDBR	17.0	17.1	18.1	17.8	17.3	17.4	16.7	16.3	16.4	16.3	16.5	16.2
		MCWBR	8.9	8.0	7.6	7.2	7.0	6.9	6.4	5.8	6.8	7.2	8.0	8.5
	5% WB	MCDBR	15.8	15.3	14.5	14.2	15.1	15.7	14.6	14.1	14.1	14.3	15.0	15.3
		MCWBR	9.0	8.1	7.6	7.4	7.1	7.2	6.4	6.3	6.8	7.1	8.6	8.8
Clear-Sky Solar Irradiance	taub		0.376	0.419	0.512	0.549	0.609	0.641	0.675	0.658	0.626	0.548	0.426	0.385
	taud		2.071	1.945	1.723	1.662	1.571	1.523	1.477	1.527	1.588	1.720	1.964	2.066
	Ebn at Noon		846	839	782	768	723	696	671	678	680	704	781	815
	Edh at Noon		140	173	230	252	276	288	301	284	260	215	155	135
All-Sky Solar Radiation	RadAvg		4.26	5.12	6.13	6.86	7.28	7.67	7.61	7.07	6.11	5.18	4.41	3.89
	RadStd		0.19	0.24	0.27	0.22	0.32	0.23	0.12	0.20	0.25	0.27	0.17	0.16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.22	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	+0.22	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page